Exercise 32.1

Is there any relationship between the probability distribution of the time taken for all trajectories to coalesce and the equilibration time of a Markov chain? Prove that there is a relationship, or find a single chain that can be realized in two different ways that have different coalescence times.

Solution:

Exercise 32.2

Imagine that Fred ignores the requirement that the random bits used at some time t, in every run from increasingly distant times T_0, must be identical, and makes a coupled-Markov-chain simulator that uses fresh random numbers every time T_0 is changed. Describe what happens if Fred applies his method to the Markov chain that is intended to sample from the uniform distribution over the states $0, 1, 2$, using the Metropolis method, driven by a random bit source as in figure 32.1b.

Solution:

Exercise 32.3

Investigate the application of perfect sampling to linear regression in Holmes and Mallick (1998) or Holmes and Denison (2002), and try to generalize it.

Solution:

Exercise 32.4

The concept of coalescence has many applications. Some surnames are more frequent than others, and some die out altogether. Make a model of this process; how long will it take until everyone has the same surname? Similarly, variability in any particular portion of the human genome is inherited like a surname. Should the fact that these surnames are subject to coalescences affect the way in which DNA fingerprint evidence is used in court?

Solution:

Exercise 32.5

How can you use a coin to create a random ranking of 3 people? Construct a solution that uses exact sampling. For example, you could apply exact sampling to a Markov chain in which the coin is repeatedly used alternately to decide whether to switch first and second, then whether to switch second and third.

Solution:

Exercise 32.6

Finding the partition function Z of a probability distribution is a difficult problem. Many Markov chain Monte Carlo methods produce valid samples from a distribution without ever finding out what Z is. Is there any probability distribution and Markov chain such that either the time taken to produce a perfect sample or the number of random bits used to create a perfect sample are related to the value of Z? Are there some situations in which the time to coalescence conveys information about Z?

Solution: